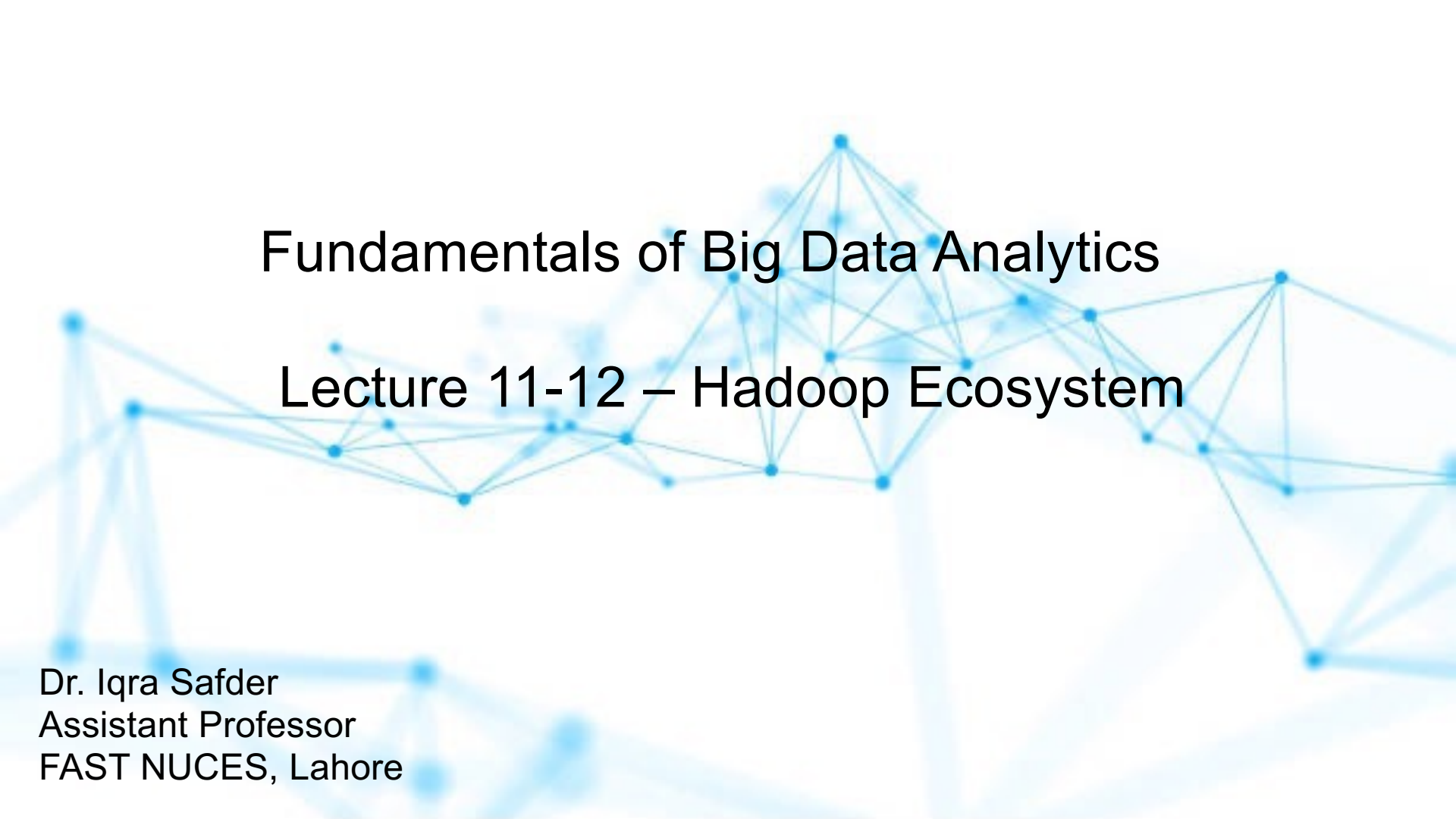


The background of the slide features a complex, abstract network diagram. It consists of numerous small blue circular nodes connected by thin, light blue lines, creating a web-like structure that spans the entire frame. The nodes are distributed unevenly, with some clusters and some isolated points, suggesting a data network or a complex system.

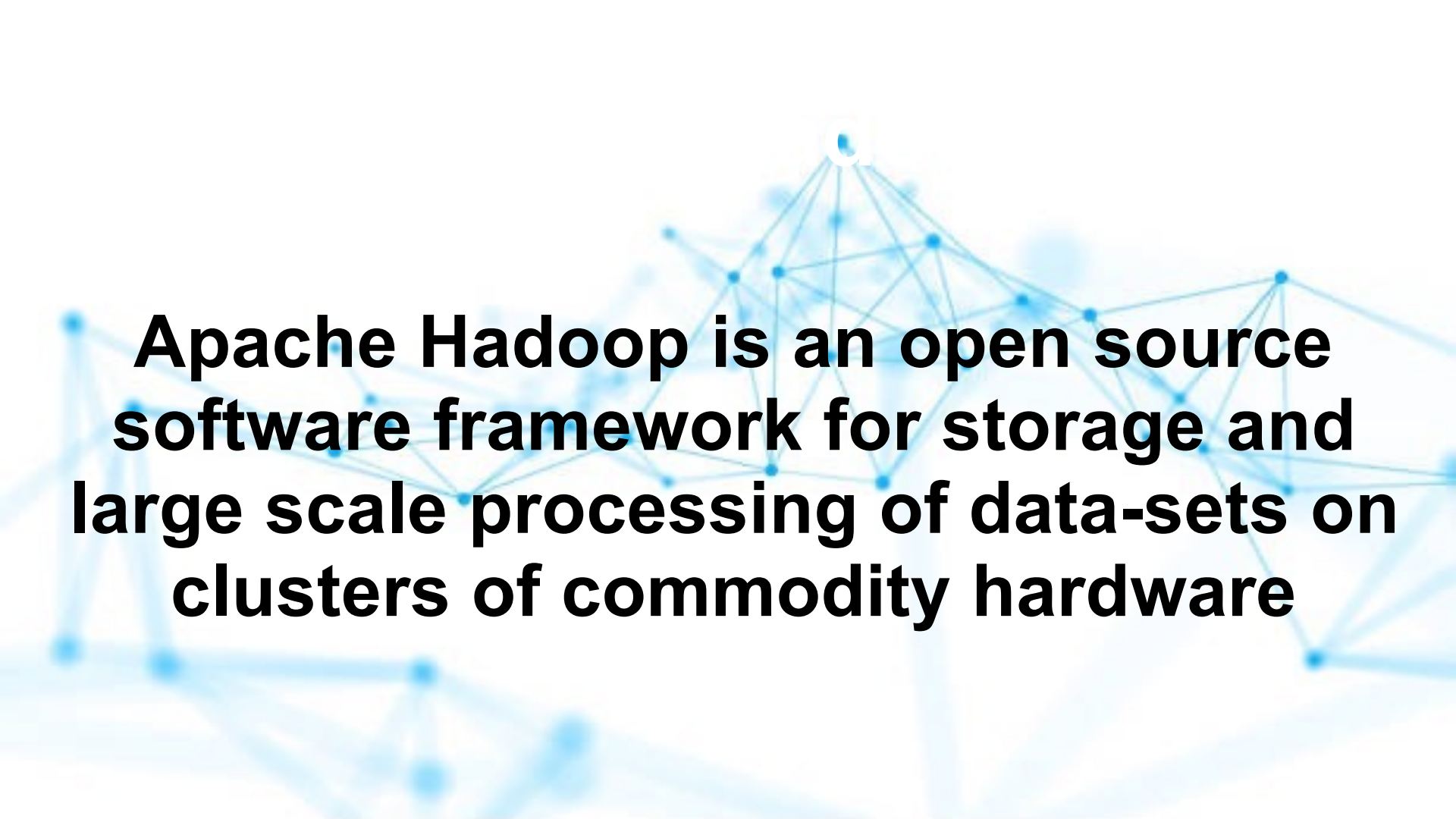
Fundamentals of Big Data Analytics

Lecture 11-12 – Hadoop Ecosystem

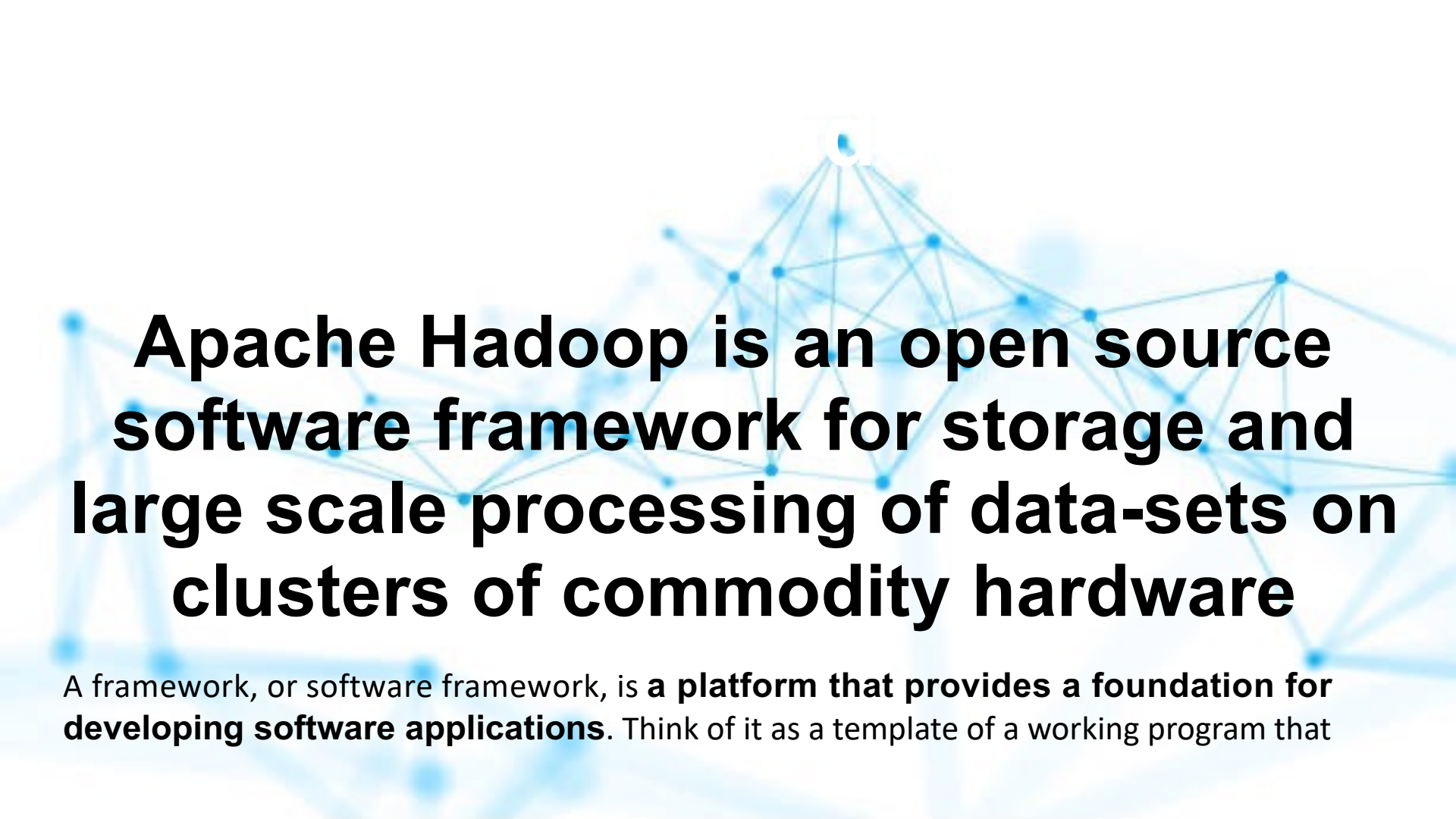
Dr. Iqra Safder
Assistant Professor
FAST NUCES, Lahore

The background of the slide features a complex, abstract network diagram. It consists of numerous small blue dots, representing nodes, which are interconnected by thin, light blue lines, representing edges. The network is distributed across the entire frame, with some areas showing denser clusters of nodes and others showing more sparse connections. The overall effect is a sense of a large-scale, interconnected system, typical of data science or network theory visualizations.

Hadoop Beginnings



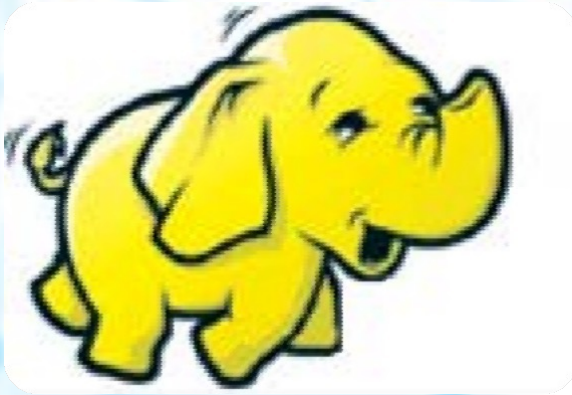
Apache Hadoop is an open source software framework for storage and large scale processing of data-sets on clusters of commodity hardware



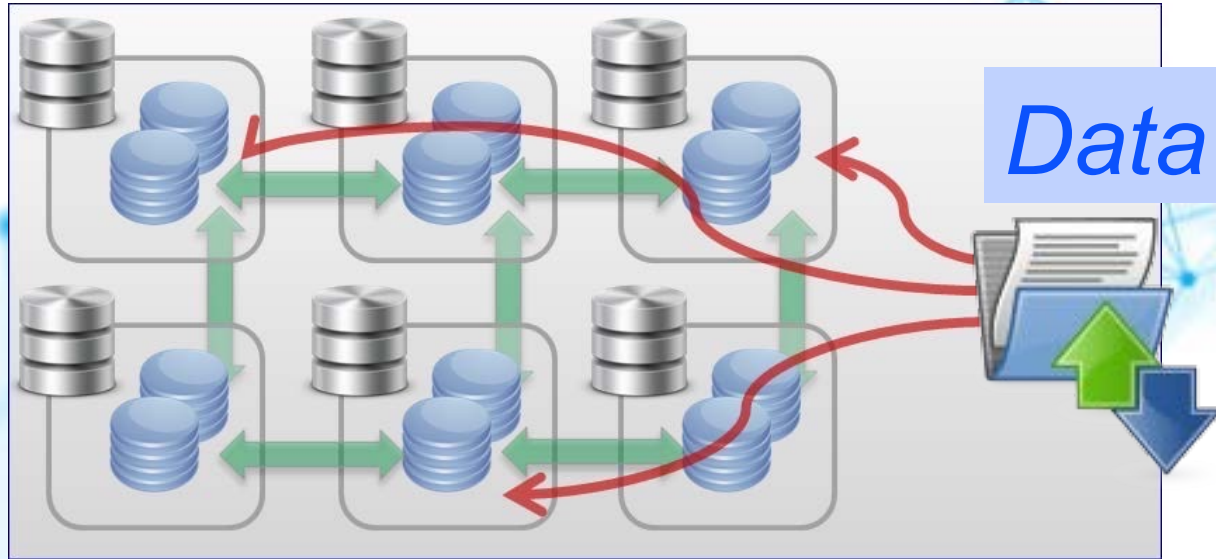
Apache Hadoop is an open source software framework for storage and large scale processing of data-sets on clusters of commodity hardware

A framework, or software framework, is **a platform that provides a foundation for developing software applications**. Think of it as a template of a working program that

**Hadoop was created by Doug Cutting
and Mike Cafarella in 2005
Named the project after son's toy
elephant**



Moving Computation to Data



Computation

Scalability at Hadoop's core!



Scalability at Hadoop's core!



- Scalability is defined as the ability to increase or decrease the size of something.
- In the case of Hadoop, this means regulating the size of the cluster itself. There are two ways to perform scalability-related operations:

more hardware.

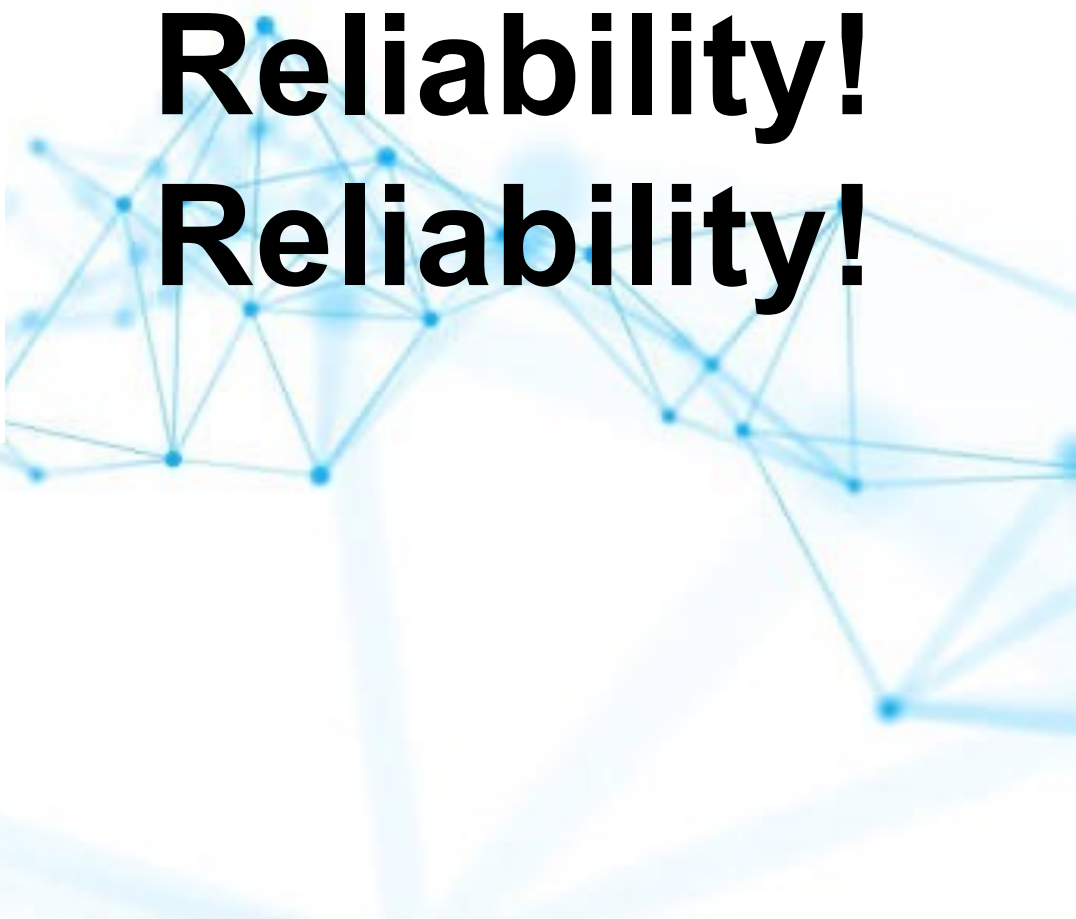


Reliability!
Reliability!



GoogleTM
BigTable

Reliability!
Reliability!



Column family

	Column family 1		Column family 2	
	<i>Column 1</i>	<i>Column 2</i>	<i>Column 1</i>	<i>Column 2</i>
Row key 1				
Row key 2				

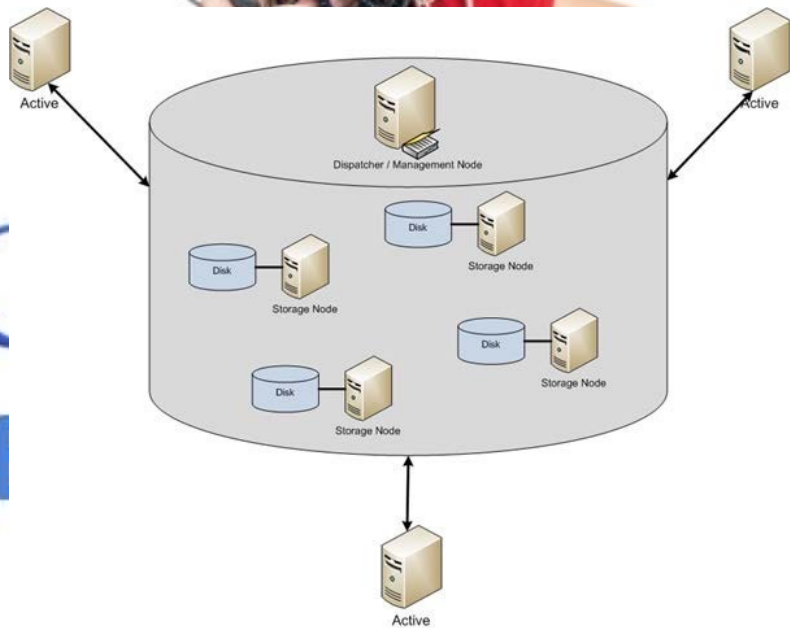
t1

t2

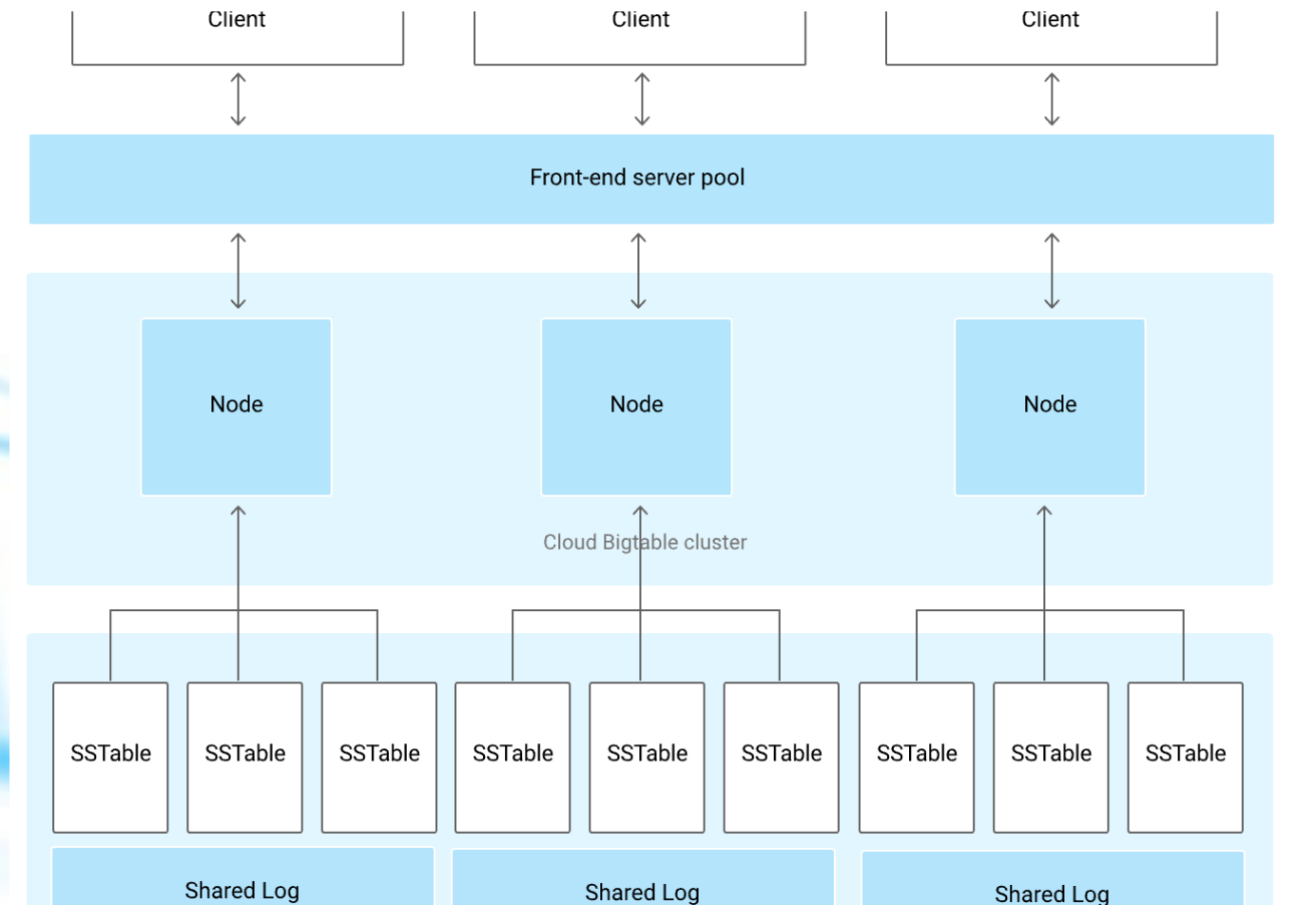
t3



Reliability! Reliability!



Google File System

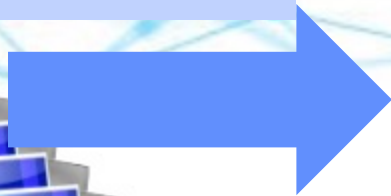




*Once
a year*



365 Computers



*Once
a day*





Hourly

New Approach to Data



New Kinds of Analysis

Schema-on read style

ANALYSIS



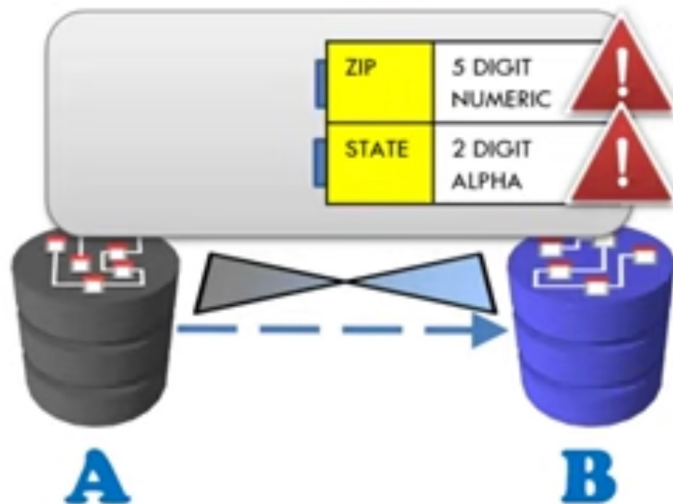
**Schema
On
Read**



HADOOP



**Schema
On
Write**



SQL

**Schema
On
Read**



A



B

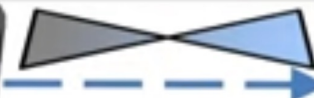
HADOOP



**Schema
On
Write**



A



B

SQL

ZIP	5 DIGIT NUMERIC
STATE	2 DIGIT ALPHA





HADOOP



SQL



 **60 SERVERS**



1000 SERVERS

HADOOP





 **60 SERVERS**



1000 SERVERS

HADOOP



SQL



 **60 SERVERS**



1000 SERVERS



JAN
FEB
MAR
APR
ETC...

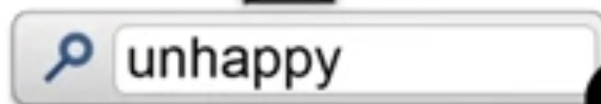


HADOOP



SQL





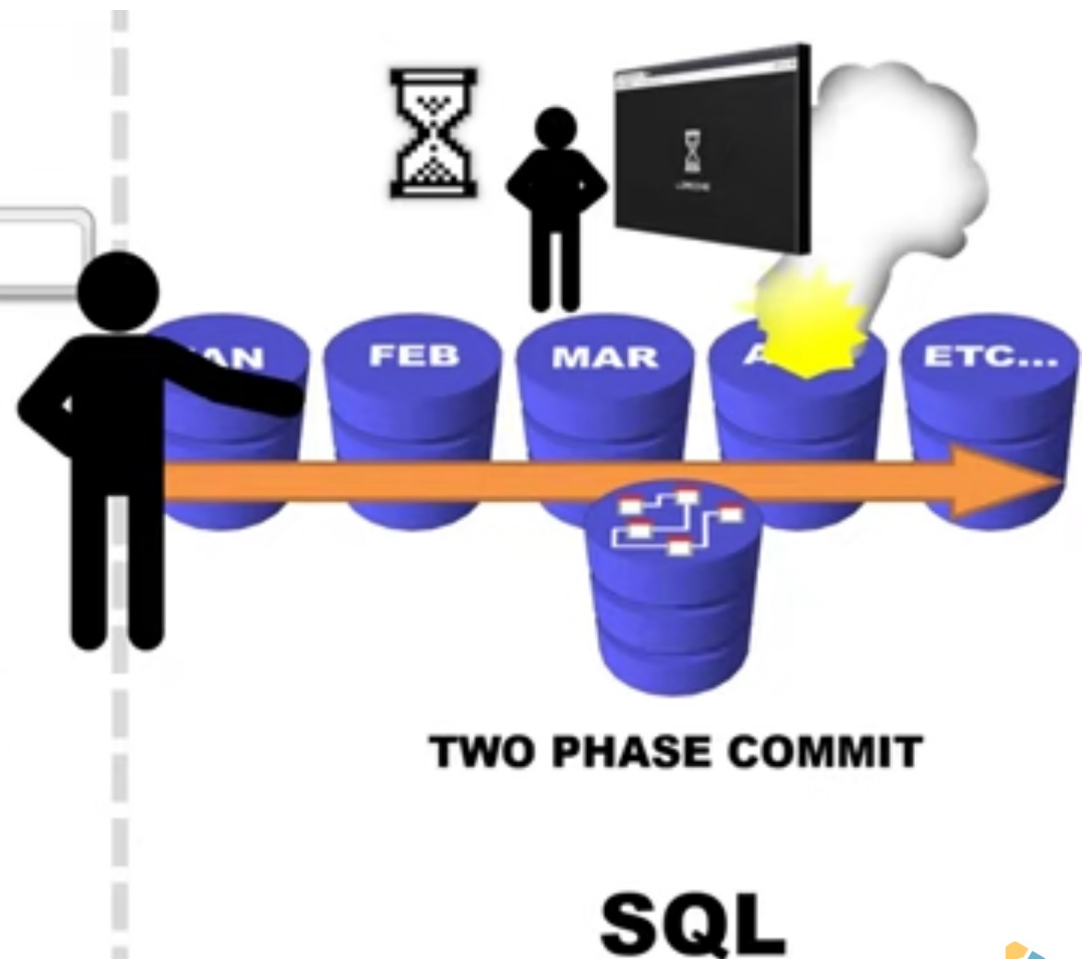
JAN
FEB
MAR
APR
ETC...



HADOOP



SQL





HADOOP



SQL



APACHE
HBASE



HADOOP

SQL

New Kinds of Analysis



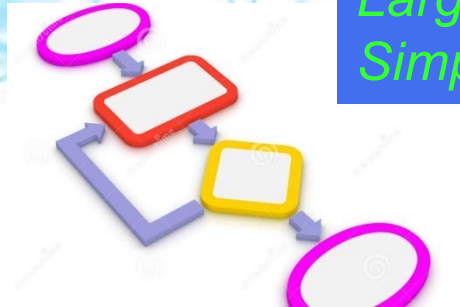
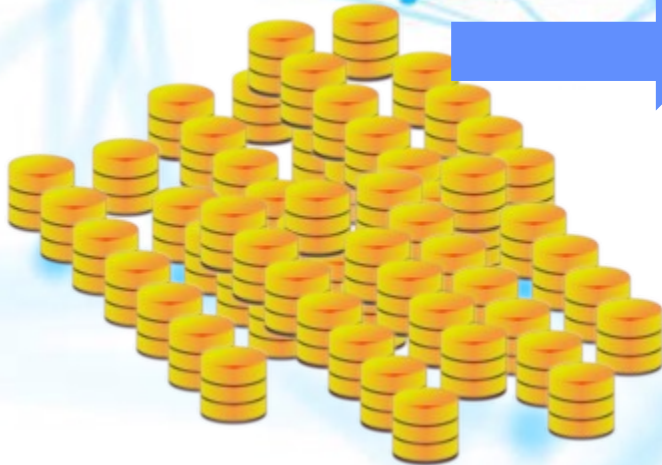
New analysis



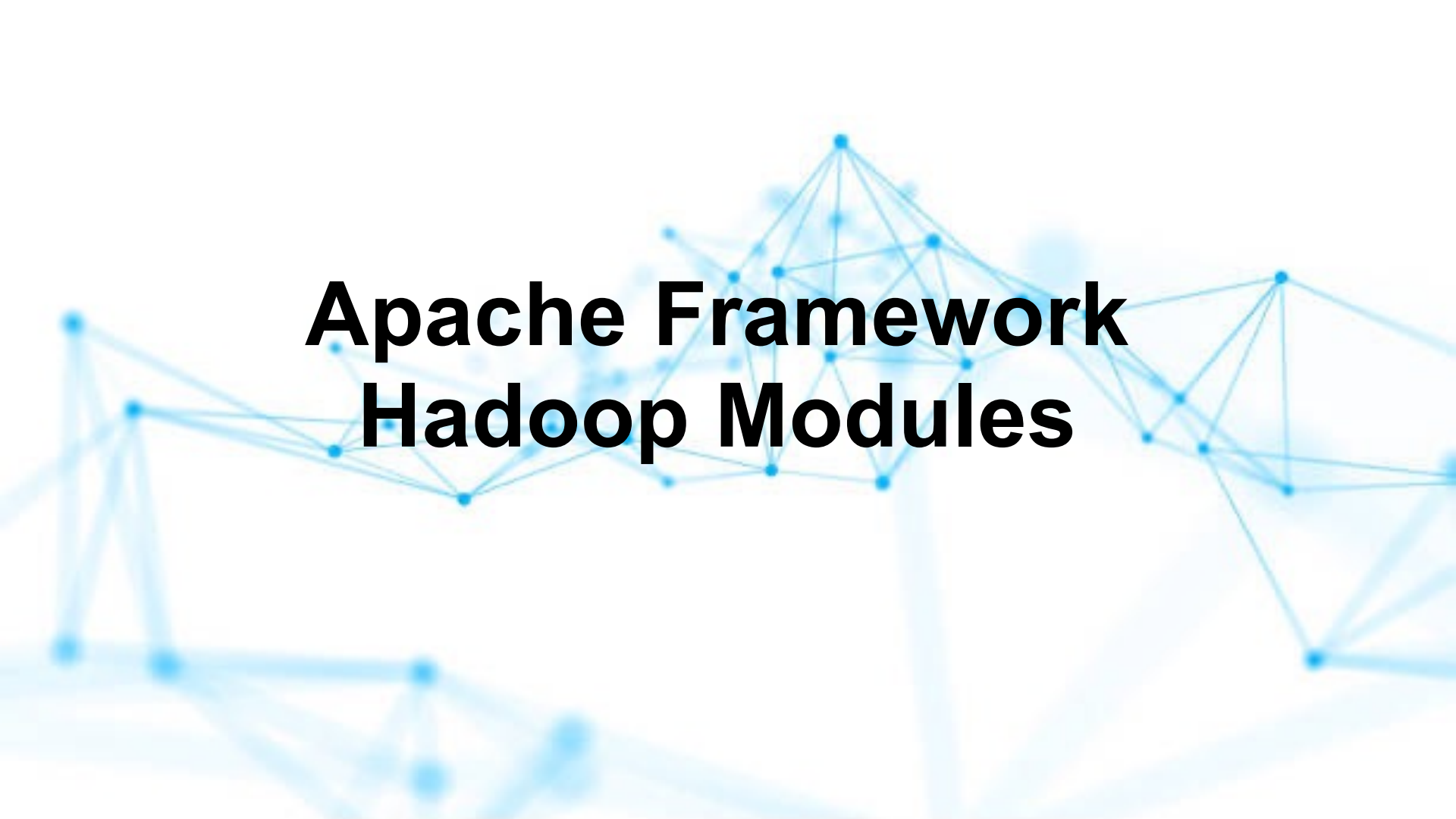


Small Data & Complex Algorithm

Vs.



Large Data & Simple Algorithm

The background of the slide features a complex, abstract network diagram. It consists of numerous small blue dots, representing nodes, which are interconnected by thin, light blue lines. These lines form a web-like structure that spans the entire width and height of the image. The nodes are distributed unevenly, with some areas having a higher density of connections than others. The overall effect is a sense of a large, interconnected system, such as a data network or a distributed computing framework.

Apache Framework Hadoop Modules

Apache Framework Basic Modules

Hadoop Common

(HDFS)

Hadoop YARN

Apache Framework

Basic Modules

Hadoop Common

Hadoop Common refers to **the collection of common utilities and libraries that support other Hadoop**

Hadoop Framework, along with the Hadoop Distributed System (HDFS), Hadoop YARN and Hadoop MapReduce.

(HDFS)

Hadoop YA R N

Apache Framework Basic Modules

Hadoop Common

**Hadoop Distributed File System
(HDFS)**

Hadoop YARN

HDFS is **a distributed file system that handles large datasets running on commodity hardware**. It is used

(and even thousands) of nodes. HDFS is one of the major components of Apache Hadoop, the others

Apache Framework Basic Modules

Hadoop Common

(HDFS)

Hadoop YARN

It's a layer that separates the resource management layer and the processing

YARN is a **resource manager** of

processing engine and the management function of MapReduce.

Apache Framework Basic Modules

Hadoop Common

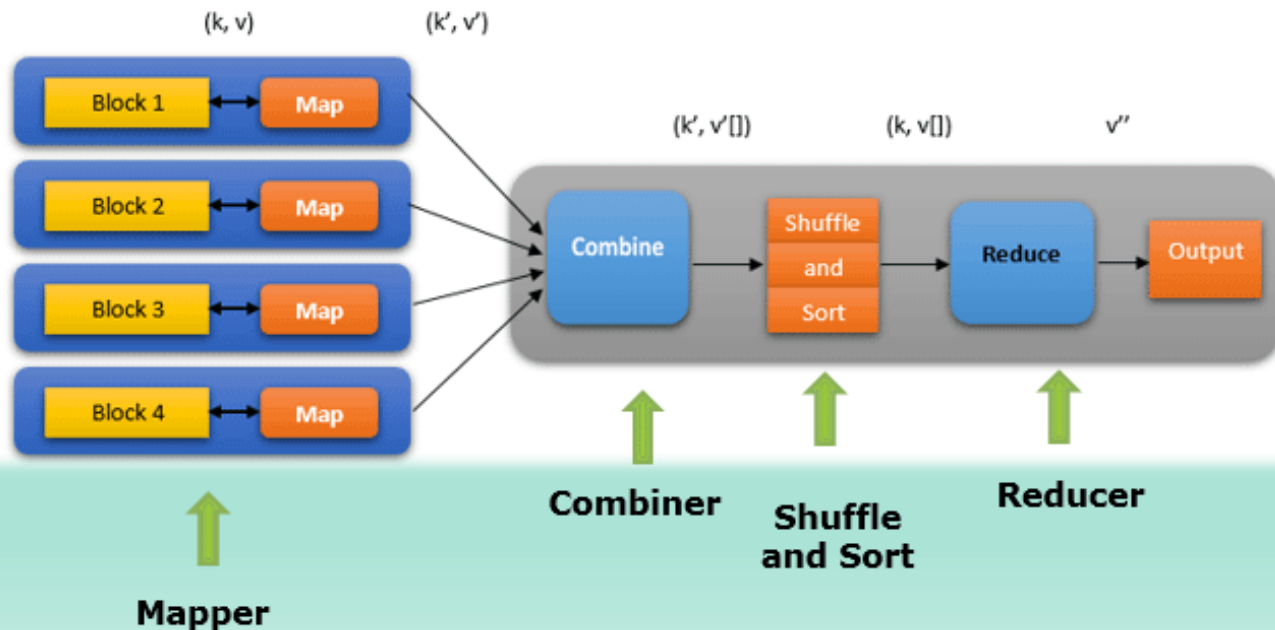
(HDFS)

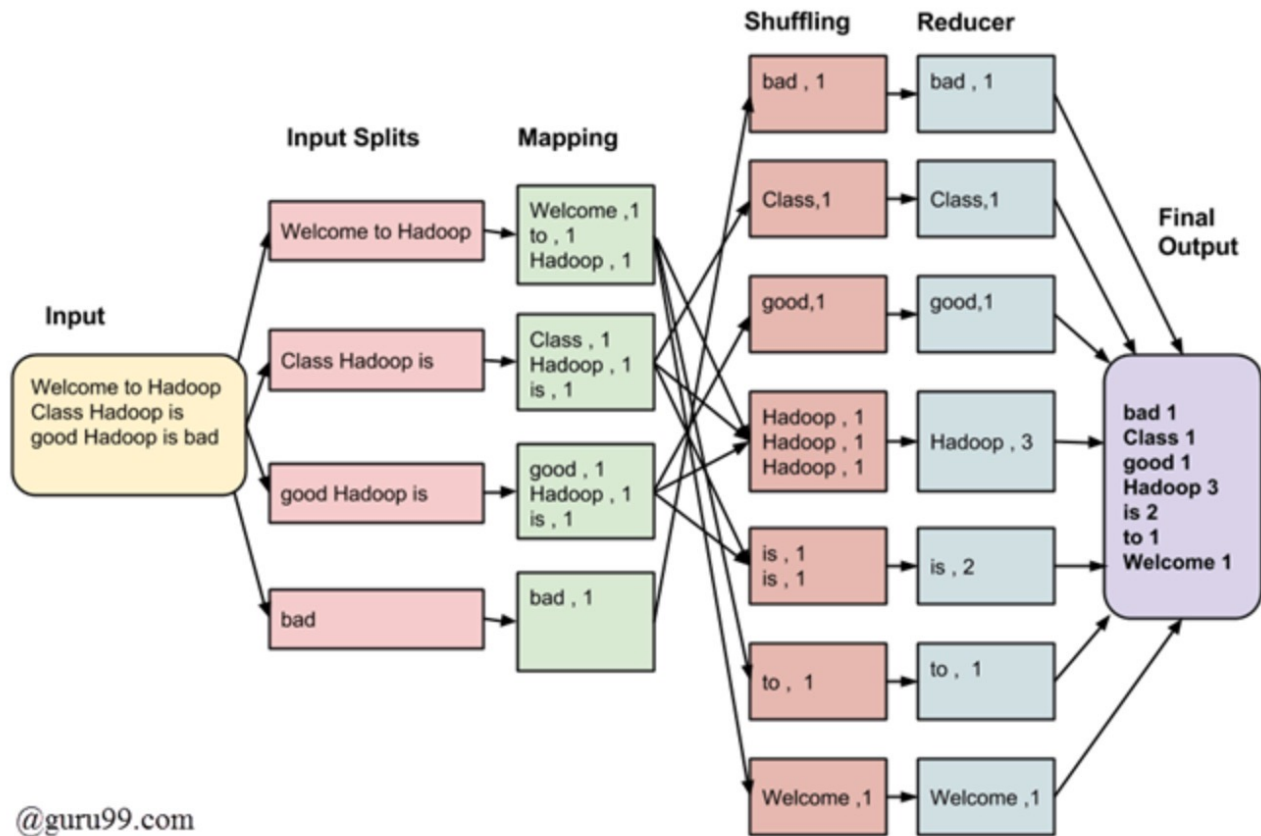
Hadoop YARN

Hadoop MapReduce

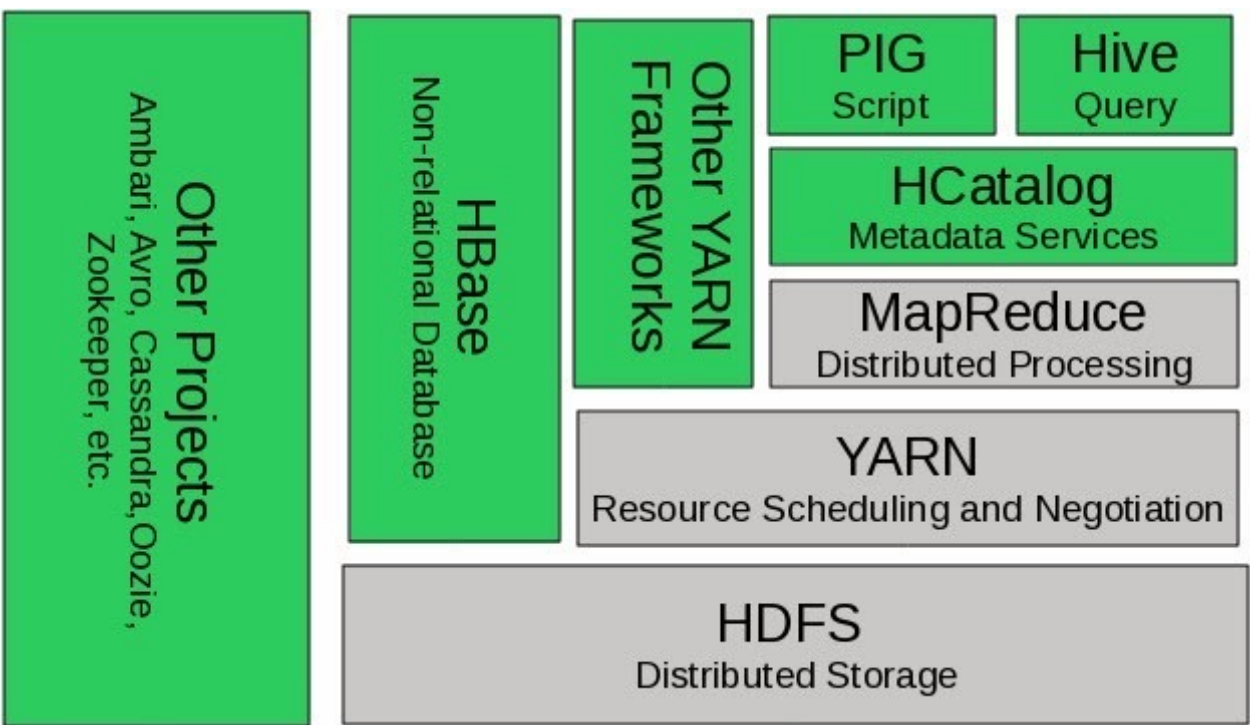
MapReduce is a Hadoop framework used for writing that can process vast amounts of data on large

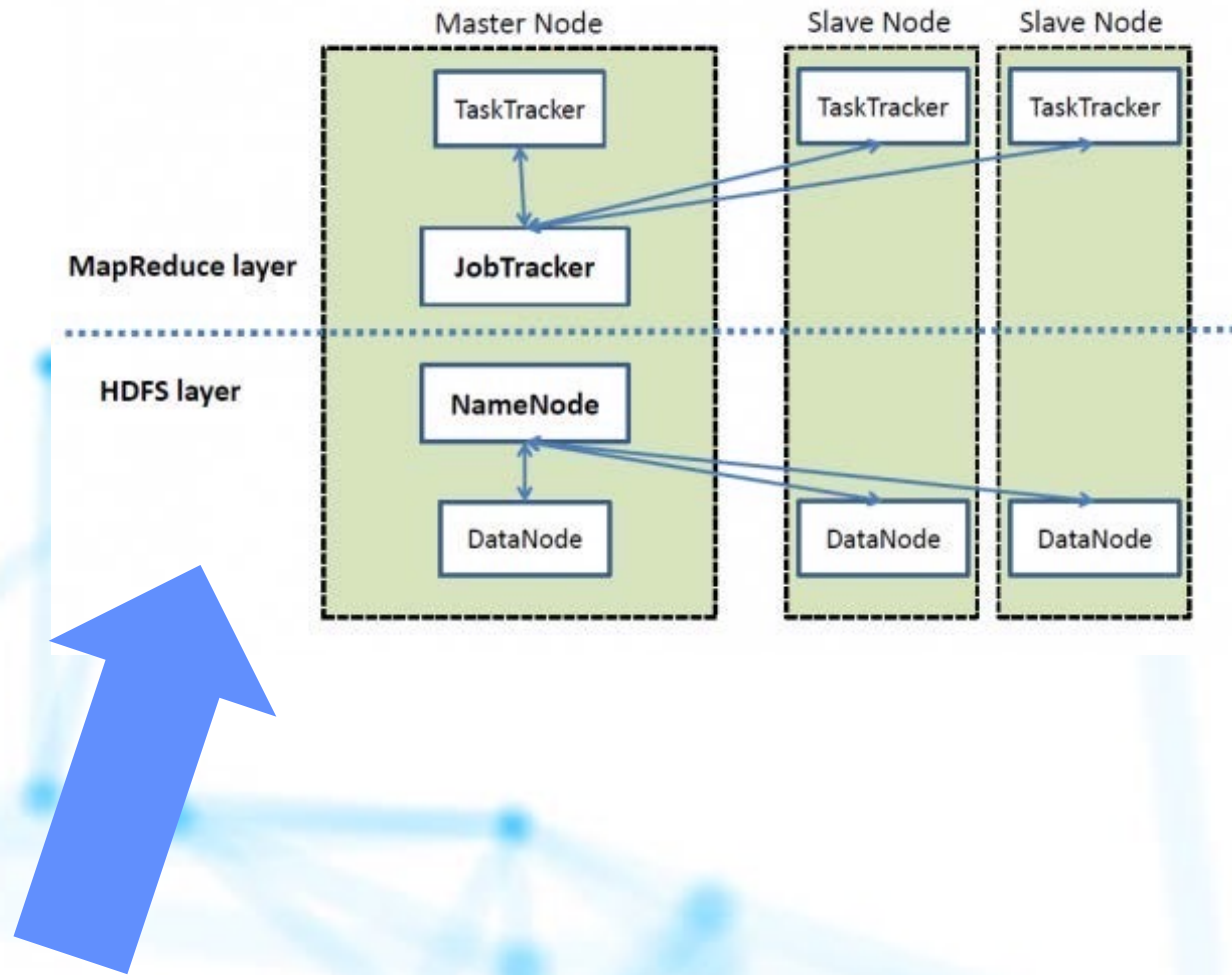
How MapReduce Works

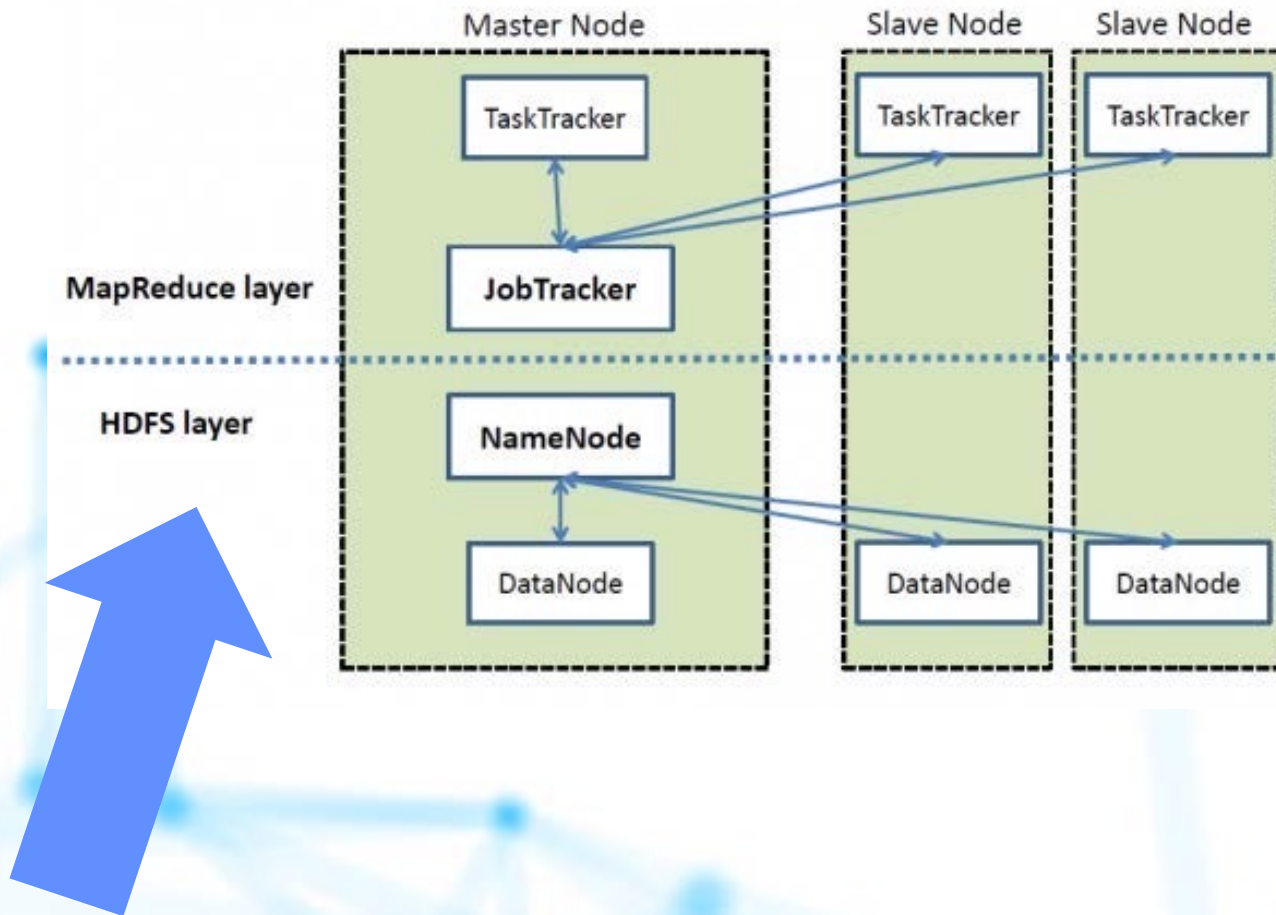




@guru99.com



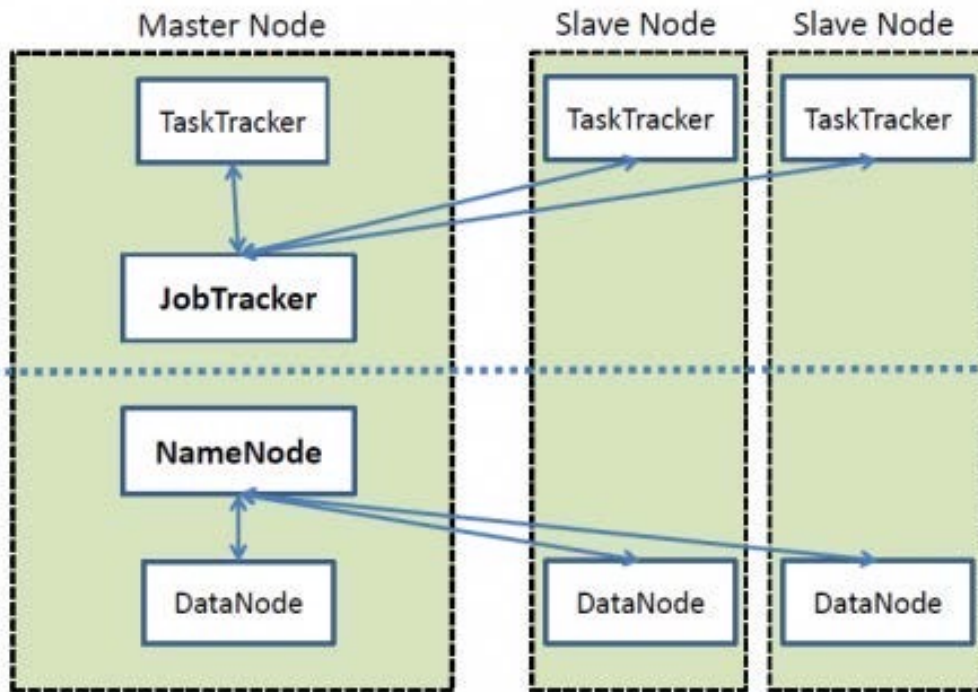


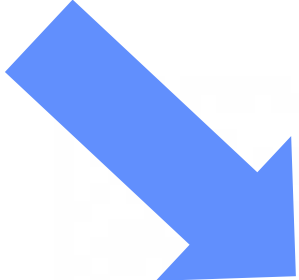




MapReduce layer

HDFS layer





MapReduce layer

HDFS layer

